Tanuj Sharma

Senior Software Engineer – Backend | Distributed Systems | Event-Driven Architecture
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SUMMARY

Backend engineer with 5+ years building event-driven payment systems in Java, Spring Boot, Kafka, and PostgreSQL on AWS. Own 12 microservices handling 1M+ transactions/day, focused on idempotency and concurrency safety.

SKILLS

Languages & Frameworks: Java 11 (Concurrency), Spring Boot, Golang, Python, SQL, REST APIs, JUnit, Mockito
Distributed Systems: Microservices, Event-Driven Architecture, Idempotency, Transactional Outbox, Caching
Data & Messaging: PostgreSQL, MySQL, Apache Kafka (Transactions, Consumer Groups, DLQ), Redis
Cloud & DevOps: AWS (ECS, RDS Aurora, Lambda, SQS, S3), Docker, Kubernetes, CI/CD, Git
Observability & AI Tooling: OpenTelemetry, Prometheus, Grafana, Resilience4j, Claude (AI-assisted development)

EXPERIENCE

Vegapay Gurugram, India
B2B credit card and lending infrastructure for banks and fintechs. Joined as the 11th employee; company now 135+.

Senior Software Engineer – Backend Apr 2025 – Present

- Run 12 production microservices at 99.99% availability and a 99.99% transaction success rate, with P99 authorization latency under 150ms.
- Cut P99 latency 65% and P50 40% on the Block Code (card risk control), transaction, and bill services using Redis cache-aside, fewer redundant calls, and async Kafka steps.
- Banks were escalating duplicate and silently dropped transactions: made APIs and Kafka consumers idempotent (DB unique constraints) and added retries, DLQ, and alerts, cutting escalations 70%.
- Fixed an EMI ledger double-posting race (sync and async Kafka write paths) seen only under production latency: moved EMI to one sync writer, reconciled affected entries, and set a single-writer rule for ledger flows.
- Drove splitting a shared commons library (used by all 13 services) into versioned domain libraries despite pushback; built the Kafka module first, trading release overhead for lower coupling (ongoing).
- Mentor 4–5 engineers; guided and reviewed their delinquency and collections module (INR 29 Cr+ collected); helped raise unit test coverage to 80%+.

Software Engineer – Backend Mar 2023 – Apr 2025

- Designed and built the core Credit Card Management System (CMS) and the Credit Line on UPI platform before the team scaled; it now handles 1M+ transactions/day and INR 1,344 Cr+ lifetime volume.
- To stop losing events during Kafka outages, designed a transactional outbox: events are saved in the same DB transaction as the business change, then published and replayed after recovery.
- Designed and built daily book closure as parallel workflows across services with optimized DB access, producing 800K+ user bills within an hour.
- Built the notification engine (SMS, email, WhatsApp) from scratch on Spring Boot, Kafka, and Redis; it sends 2M+ messages/day for 5 issuing banks.

BharatPe Gurugram, India
Software Developer Jul 2021 – Mar 2023

- Built BharatPe Card and PostPe card lifecycle features (1M+ active accounts): onboarding, activation, spend limits.
- Optimized onboarding and authorization queries and indexes, cutting latency from 0.9–1.5s to under 400ms.
- Customers were stalling between onboarding stages; debugged the failures and restructured the flow logic, getting 45% of stuck customers fully onboarded.

Grab Remote / Bangalore, India
Software Engineer Intern Feb 2021 – Jul 2021

- Built a concurrent Golang diagnostic service that detects failovers and connection-pool exhaustion across MySQL, PostgreSQL, and MSSQL, cutting developer MTTR by 80%.

EDUCATION

B.Tech, Computer Science & Engineering, Poornima College of Engineering, Jaipur 2017 – 2021